Directional Prediction from Non-Tumorous Tissue: Localizing Breast Tumors in MRI Using CNNs

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1 Introduction

Breast cancer detection remains challenging, especially in dense breasts (Abeelh and AbuAbeileh, 2024). While MRI is highly sensitive, it suffers from high false-positive rates (Abeelh and AbuAbeileh, 2024; Oviedo et al., 2025). Most deep learning approaches focus on directly visible tumor regions (Fujioka et al., 2024). However, growing evidence suggests non-tumorous tissue contains diagnostically relevant information (Li et al., 2025; Baughan et al., 2023), potentially due to field cancerization, where normal cells are replaced with with cancer-primed cells (Curtius et al., 2017; Lochhead et al., 2014). This process seems to eventually lead to changes in imaging features, as it was found that tissue closer to a tumor is more similar to the tumor itself than tissue further away (Baughan et al., 2023). Based on this finding, this study investigates whether a convolutional neural network (CNN) can extract directional cues from non-tumorous tissue in MRI patches. Specifically:
(1) Can a 2D CNN, trained on 96×96 pixel tumor-free MRI patches, predict the unit vector pointing toward the tumor centroid in single focal breast cancer cases, achieving a median angular error significantly smaller than a randomized control model?
(2) Do the characteristics of the CNN's saliency maps vary systematically with tumor subtype and distance from the tumor?

The results showed that a CNN trained on tumor-free MRI patches, predicted tumor direction with a median angular error of 31.8°, outperforming a randomized baseline (69.7°) significantly. Saliency analysis revealed that the model learns subtype- and distance-specific tissue signatures; these signatures alone allow perfect classification of tumor subtype and distance group and explain 85% of the variance in prediction errors.

1.1 Related Work

Deep learning has been widely implemented in breast cancer research. Applications include classification, segmentation, outcome and treatment-response prediction (Fujioka et al., 2024). Breast segmentation models for example, are able to reach high accuracy with Dice Scores of up to 95% (Zheng et al., 2025). Architecture combining segmentation and classification reported up to 90% accuracy (Jaincy and Pattabiraman, 2025).
However, as many detection strategies are based on the differentiation between normal and tumorous tissue(Gadaleta et al., 2022; Abeelh and AbuAbeileh, 2024), only few models are able to detect cancer in its early stages. Growing evidence suggests that non-tumorous tissue contains diagnostically relevant information (Li et al., 2025, 2023; Baughan et al., 2023), which could be utilized to improve early stage detection. Field cancerization leads to cellular and molecular alterations that predispose the region for abnormal tissue growth (Lochhead et al., 2014) and is known to vary by tumor subtype (Bedrosian et al., 2020). The effects of field cancerisation could explain why a study by Li et al. (2023) indicated that non-cancerous breast tissue has some predictive power of future malignancy in the contralateral breast.
In fact, CNN trained by including "normal" tumor-surrounding tissue showed increased accuracy, demonstrating that peritumoral regions contain recognizable cancer-specific features (Li et al., 2025). Saliency maps often highlight regions outside the lesion (Pertuz et al., 2023) and tissue close to a tumor is more similar to the tumor than tissue further away (Baughan

et al., 2023).
A study by Oviedo et al. (2025) localized hard-to-detect tumors by developing a CNN that learns begin breast tissue characteristics and marks deviations as suspicious areas.
Although the approach is similar to utilizing field cancerization characteristics, the model still relies on the presence of some tumorous tissue.
This research conceptually extends Oviedo et al. (2025) by investigating whether the "normal" tissue surrounding a tumor can be used to infer its location. Unlike Oviedo et al.'s approach, which learns benign characteristics to flag anomalies, this study method aims to leverage malignancy-associated markers within tissue.
This could address a key limitation of Oviedo et al. (2025) model, which struggled when the tumor-tissue boundary was ambiguous: in small or low-enhancement lesions (minimal signal or low contrast) and in non-mass or large, diffuse tumors (poorly defined margins). By relying on directional signals in non-tumorous tissue, issues caused by a lack of clear tumor boundary might be circumvented. In addition, the characteristics of the malignant tissue might even provide more diagnostic information, as field cancerization is known to vary by tumor subtype (Bedrosian et al., 2020).
This proof-of-concept study tests whether directional information can be extracted from non-tumorous MRI tissue. To interpret the model's decisions, saliency mapping is employed. Saliency maps are usually used to link features to known predisposed anatomy or biomarkers (Brima and Atemkeng, 2024), here it is probed whether the CNN's attention varies systematically with tumor distance and subtype. This would indicate that the model learned from the heterogeneous field effects (Bedrosian et al., 2020; Wang et al., 2025).
Angular regression is more commonly used for anatomic landmark localization (Noothout et al., 2020) and is often evaluated based on distance based errors (Cai et al., 2020; Noothout et al., 2020).
Given the goal of investigating the existence of directional cues, the use of distance-based error benchmarks was considered unsuitable. Instead, performance is validated against a chance-level baseline. The presence of such directional cues would demonstrate the general ability to infer "directions of increased suspicion" from non-tumorous tissue, providing a basis to integrate the approach into spatial anomaly detection methods in the future.

2 Methods

2.1 Data

The Mama Mia dataset was utilized. It contains dynamic contrast-enhanced 3D MRI scans with expert tumor segmentation of different tumor subtypes. Only the 807 single focal cases (i.e., one tumor is present) were selected to ensure that the directional signal in each patch is only attributed to one tumor.

2.2 Patch Extraction and Pre-processing

Only tumor containing slices are selected. To reduce spatial autocorrelation, patches were sampled from every third slice. From each selected slice, 96 x 96 pixel images were extracted. The patches were sampled according to the following requirements: (1) complete exclusion of the tumor and a 15 mm surrounding zone, (2) maximum distance of 120 mm from the tumor boundary, (3) a maximum overlap of 65% of the x-y coordinates, (4) a minimum of 65% of the breast-tissue pixels.
Breast tissue was segmented from the background using a U-Net-based model, MRI_BreastSeg (FloXyPython, n.d.; Ronneberger et al., 2024). This resulted in a total of 3303 patches.
For each patch, a 2D vector was calculated from the patch center to the centroid of the tumor in the same slice. The tumor centroid was used as the target as it provides a stable reference across patches, reducing variability caused by irregular tumor shapes. This vector was normalized to a unit vector, focusing the learning objective purely on direction rather than distance. The rationale is to reduce sensitivity to participant specific factors (e.g., tissue density, breast and torso shape). A combined prediction was avoided to prevent optimization of one prediction at the expense of the other.
The data was split into training (70%), validation (15%), and test (15%) sets, with strict patient separation.

2.3 Model

A custom 2D CNN was implemented for directional regression. The architecture (detailed in Appendix **??**) consisted of four convolutional blocks with increasing filter counts (32, 64, 128, 256), each followed by batch normalization and max-pooling. The output of a global average pooling layer was passed into three fully connected layers (512, 256, 128 units) with batch normalization and dropout(0.4, 0.3, 0.2). The final two-unit linear layer outputs the predicted directional vector components. The model was trained using the Adam optimizer and a cosine loss function.

2.4 Performance Evaluation

Model performance was evaluated using absolute angular error between the predicted and ground-truth vectors. Descriptive statistics were calculated. To establish a comparison with a randomized control model an identical CNN architecture was trained on the dataset with scrambled prediction targets. To assess statistical significance between the model and randomized control a paired t-test and Wilcoxon signed-rank test were performed.

Performance was further analyzed by tumor subtype and distance. The distance from patch-center to tumor centroid was binned into 15 mm intervals. Group differences were evaluated using two-way ANOVA. Effect sizes were calculated using η_p^2). Additionally, the predictive power of subtype and distance on individual errors was assessed using linear regression.

2.5 GradCam Evaluation

Grad-CAM was applied to the fourth convolutional layer. For each resulting saliency map spatial entropy (i.e., dispersion of attention), spatial sparsity (i.e., concentration of attention), and radial ratio (i.e., centrality of attention relative to the patch center) were calculated.

The influence of subtype and distance on these saliency metrics was analyzed using two-way ANOVA. To investigate whether these saliency metrics encode tumor information Random Forest classification was used to predict subtype and distance class together.

Lastly, the relationship between saliency metrics and the model's angular prediction error was investigated using Random Forest regression.

3 Results

3.1 Performance Evaluation

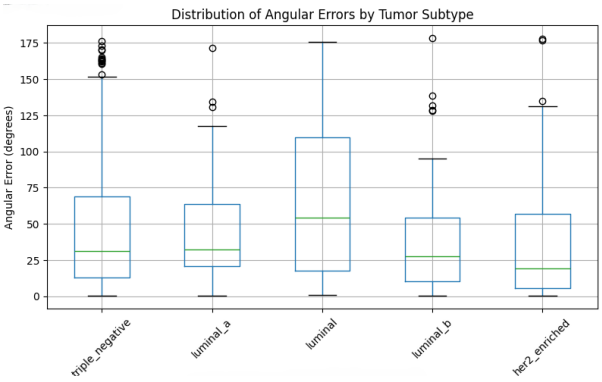


Figure 1: Angular prediction error by tumor molecular subtype. HER2-enriched showed the lowest median error (19.2°), while Luminal showed the highest (54.2°). HER2-pure excluded due to low test set representation.

The model achieved a median angular error of 31.8° (mean 48.2°, SD 45.4°), with 60.0% of predictions within 45° and 48.9% within 30°. Thus, it significantly outperformed the randomized control model (median 69.7°, mean 76.3°, SD 51.4°; Wilcoxon signed-rank test $p < 0.001$; paired t-test: $t = -9.51$, $p < 0.001$), indicating that CNNs can learn directional cues from non-tumorous tissue (RQ1).

In addition, performance varied by subtype and distance. The HER2-enriched subtype showed the lowest median error (19.2°), while Luminal showed the highest (54.2°) (Table 1, Appendix A.2). Error generally increased with distance (Table 2, Appendix A.2), with the lowest median error (25.4°) in the 45–60mm bin and highest (52.5°) in the 90–120mm bin.

A two-way ANOVA revealed a significant main effect for tumor subtype ($F=10.93$, $p=0.001$, $\eta_p^2 = 0.105$) but not for distance ($p=0.105$) (Table 3, Appendix A.3). However, subtype and distance explained negligible variance of individual angular errors (linear regression $R^2 = -0.039$), indicating high within-group variability.

Angular error mostly increased with distance from the tumor. The lowest median error (25.39°) was observed in the 45-60mm bin, while the highest median error (52.50°) occurred in the 90-120mm bin. However, error did not appear to increase monotonically with distance; as

the 60-90mm bin showed a lower median error (29.96°) than the closer 15-30mm bin (36.35°).

3.2 GradCam Evaluation

Saliency maps revealed that the model's visual attention varied by both tumor subtype and distance. Two-way ANOVA showed significant main and interaction effects for spatial entropy and radial ratio (all $p < 0.001$; see Table 4 in Appendix A.4), indicating that the CNN learned distinct patterns. To test whether the saliency maps encoded information about the tumor Random Forest Classification was used. The classifier achieved 100% accuracy in predicting the tumor subtype and distance bin combined using the three saliency metrics. This is a strong indication that the CNN learned distinct patterns corresponding to subtype specific field cancerization effects (RQ2). Furthermore, a Random Forest regressor using these same features explained 85% of the variance ($R^2 = 0.853$) in the model's angular prediction error, with spatial entropy being the most important predictor (importance = 0.414), followed by radial ratio (0.388) and spatial sparsity (0.199).

4 Discussion

This study provides the first proof-of-concept that a CNN can extract directional signals from non-tumorous breast MRI tissue, achieving a median angular error significantly better than chance (31.8° vs. 69.7°, $p < 0.001$). More importantly, the model's saliency map characteristics, varied systematically with tumor subtype and distance. This suggests that the CNN learned tumor subtype-specific tissue changes, as have been described in field cancerization and previous CNN models (Bedrosian et al., 2020; Wang et al., 2025).

While the directional accuracy does not allow clinical use, the results establish that peritumoral tissue contains directional information. These could complement existing detection models, such as the anomaly detection model by Oviedo et al. (2025). Integrating an anomaly detection system based on peritumoral tissue could create a framework capable of correcting misclassifications caused by ambiguous tumor boundaries. Unlike models that localize visible tumors, the method in this paper points from lesion-free regions toward tumors, potentially aiding early-stage cancer detection.

Notably, saliency map characteristics alone allowed perfect classification of tumor subtype and distance group via Random Forest, revealing that the CNN learned subtype-specific tissue signatures without explicit supervision.

These saliency features also strongly predicted angular error ($R^2 = 0.85$), suggesting they could serve for confidence estimation of individual predictions. Confidence estimations could allow to weight the loss during training (similar to focal loss) to handle the variance in the training example difficulty.

The current model could be used to iteratively process an MRI slice, estimating the direction and saliency based distance estimation and accumulating each estimate to predict either tumor location or distance heatmap. This would be conceptually similar to anomaly-detection but would operate by highlighting regions around the tumor before it becomes visible.

Future iterations of the model could perform anomaly detection more directly by being trained on entire tumor-free slices close to the tumor.

The study has several limitations. (1) the study does not contain a healthy control group, (2), the use of 2D patches ignores three dimensional information about the tissue (3) the use of only single-focal tumors might not generalize well to multi-focal cases, (4) the prediction of the direction to the tumor centroid neglects the possibility that irregularly shaped tumors elicit more diffuse effects. Lastly, it is unclear whether the signal that is utilized is actually caused by the tumor, or by the field cancerization that preceded it. Studies such as Baughan et al. (2023) show similarity between tumors and adjacent tissue, but whether this reflects tumor-induced changes or pre-existing predisposition is unclear. Future research could address this question by using longitudinal study data, which would advance our understanding of the mechanisms that lead from field cancerization to tumor development, as advocated by Gadaleta et al. (2022).

In conclusion, this study demonstrated that CNNs can extract directional and subtype-specific signals from non-tumorous breast MRI tissue. These results point to a new, complementary approach of anomaly detection that does not rely on the presence of visible tumors.

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A Appendix

A.1 Appendix One

Layer (type)	Output Shape	Param #
conv1 (Conv2D)	(None, 96, 96, 32)	320
bn1 (BatchNormalization)	(None, 96, 96, 32)	128
pool1 (MaxPooling2D)	(None, 48, 48, 32)	0
conv2 (Conv2D)	(None, 48, 48, 64)	18,496
bn2 (BatchNormalization)	(None, 48, 48, 64)	256
pool2 (MaxPooling2D)	(None, 24, 24, 64)	0
conv3 (Conv2D)	(None, 24, 24, 128)	73,856
bn3 (BatchNormalization)	(None, 24, 24, 128)	512
pool3 (MaxPooling2D)	(None, 12, 12, 128)	0
conv4 (Conv2D)	(None, 12, 12, 256)	295,168
bn4 (BatchNormalization)	(None, 12, 12, 256)	1,024
gap (GlobalAveragePooling2D)	(None, 256)	0
dense1 (Dense)	(None, 512)	131,584
bn5 (BatchNormalization)	(None, 512)	2,048
dropout1 (Dropout)	(None, 512)	0
dense2 (Dense)	(None, 256)	131,328
bn6 (BatchNormalization)	(None, 256)	1,024
dropout2 (Dropout)	(None, 256)	0
dense3 (Dense)	(None, 128)	32,896
bn7 (BatchNormalization)	(None, 128)	512
dropout3 (Dropout)	(None, 128)	0
output (Dense)	(None, 2)	258

Total params: 689,410 (2.63 MB)
Trainable params: 686,658 (2.62 MB)
Non-trainable params: 2,752 (10.75 KB)

Figure 2: Architecture of used CNN.

A.2 Appendix Two

Table 1: Model performance by tumour subtype

Subtype	Median (°)	Mean (°)	SD (°)
HER2-enriched	19.2	39.88	48.57
Luminal B	27.4	39.71	39.80
Luminal A	32.1	43.65	36.08
Triple-negative	31.1	48.99	46.36
Luminal	54.2	63.95	51.17
Overall	**31.79**	**48.19**	**45.40**

Table 2: Model performance by distance from tumor

Distance Bin	Median (°)	Mean (°)	SD (°)
15-30mm	36.35	51.25	45.76
30-45mm	30.34	44.82	40.76
45-60mm	25.39	33.82	33.53
60-90mm	29.96	48.47	47.16
90-120mm	52.50	63.13	57.03

A.3 Appendix Three

Table 3: Two-way ANOVA results for angular error in tumor direction prediction

Source	F-statistic	p-value	Partial η^2
Tumor Subtype	10.93	0.0010	0.105
Distance Group	2.64	0.1047	0.022
Subtype × Distance Interaction	1.42	0.1238	0.057

A.4 Appendix Four

Table 4: Statistical analysis of Grad-CAM saliency metrics for all predictions $\leq 180°$

Metric	Effect	F-Effect	p-value	η^2p
Spatial Entropy	Tumor Subtype	4.03	0.0014**	0.040 (S)
	Distance Group	0.53	0.7122	0.004 (N)
	Interaction	5.61	0.0000***	0.194 (L)
Spatial Sparsity	Tumor Subtype	1.42	0.2151	0.014 (S)
	Distance Group	1.17	0.3239	0.009 (N)
	Interaction	1.67	0.0530	0.067 (S)
Radial Ratio	Tumor Subtype	9.90	0.0000***	0.093 (M)
	Distance Group	0.68	0.6042	0.006 (N)
	Interaction	2.24	0.0048**	0.088 (M)